

STRUCTURAL STUDIES OF DIPTERA AND LEPIDOPTERA PENTAMERIC RECEPTOR FOR THE DESIGN OF INSECTICIDAL PROTEINS

Farah FOULADKAR^a, Lefteris ZARKADAS^b, Delphine BAUD^a, Céline JUILLAN-BINARD^a, Guy SCHOEHN^b, Corinne DENIAUD^a, Hugues NURY^a

a Institut de Biologie Structural, IBS, UGA, CNRS, UMR5075, Grenoble , France

b Univ. Grenoble Alpes, CNRS, CEA, EMBL, ISBG, 71 avenue des Martyrs, Grenoble, France

(corresponding author: farah.fouladkar@ibs.fr)

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ABSTRACT

Farmers use insecticides to protect their crops. However, these products cause environmental problems due to their lack of selectivity: while pests are killed, beneficial insects such as bees are also harmed. Can scientists contribute to developing safer strategies for crop protection? In our laboratory, we focus on insect pentameric ligand-gated ion channels (pLGICs). These membrane receptors consist of five subunits, each containing three distinct domains: an extracellular domain housing the neurotransmitter binding site, a transmembrane domain with four alpha helices, and a disordered intracellular domain (1). pLGICs play a prominent role in fast neurotransmission, with receptors sensitive to acetylcholine, GABA or glutamate. Neuronal pLGICs also form the largest class of targets of insecticides (2). A mysterious, understudied, subclass of insect pLGICs is found not in the nervous system but in the gut, known as clade-1 receptors.

We set up to shed light on the molecular mechanism of those gut receptors, and on their potential as insecticide targets. Clade-1 receptor, first identified in *Drosophila* and known as HODOR, acts as a sensor essential for larval growth and development (3). Despite its biological importance, HODOR remains poorly characterized. Three HODOR homologues are present in *Drosophila*, and a single homologue has been identified in mosquitoes and lepidoptera. To investigate this receptor family, we cloned several HODOR homologues from dipteran and lepidopteran species and expressed them in Sf21 insect cells. Protein expression was monitored by flow cytometry and fluorescence-detection size-exclusion chromatography (FSEC). Preliminary results obtained for the mosquito and *Plutella xylostella* HODOR homologues are highly encouraging. We successfully purified both receptors with high yields and good stability over a range of pH conditions. Initial cryo-electron microscopy studies have yielded a high resolution structure for these receptors. Our next objective is the development of high-affinity binders targeting Clade-1 receptors, which will subsequently be screened using a yeast display platform to identify candidates with selective insecticidal activity.

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